Jahangirnagar University
জাহাঙ্গীরনগর বিশ্ববিদ্যালয়

Department of Computer Science and Engineering

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2nd Year 1st Semester (Hons)

Course Title: Digital Logic Design Laboratory

Course Code: CSE-254

Experiment No: 02

Experiment Title: Implementation of the boolean expression $Y = \bar{A}BC + A\bar{B}C + AB\bar{C}$ using logic gates.

Submitted To:


Dr. Jugal Krishna Das, B.Sc(Donetsk), MSc.(Donetsk), PhD(Kiev)

Professor,
Department of Computer Science and Engineering,
Jahangirnagar University, Savar, Dhaka-1340.

Nadia Afrin Ritu, B.Sc (JU), MSc. (JU)

~~Lecturer~~, Assistant Professor,
Department of Computer Science and Engineering,
Jahangirnagar University, Savar, Dhaka-1340.


29.8.25

Submitted By:


| | |
|---|---|
| Name: | Murad Hossain Rohan |
| Roll: | 338 |
| Registration: | 20230657073 |
| Session: | 2022-23 |
| Exam Roll: | 231041 |
| Batch: | 52 |

## Experiment No: 2

**Experiment Name:** Implementation of the boolean expression $Y = \bar{A}BC + A\bar{B}C + AB\bar{C}$ using logic gates.

## Objectives:

- To implemen boolean expression using logic gates circuits
- To verify the truth table experimentally
- To understand the practical realization of boolean algebras in digital circuits

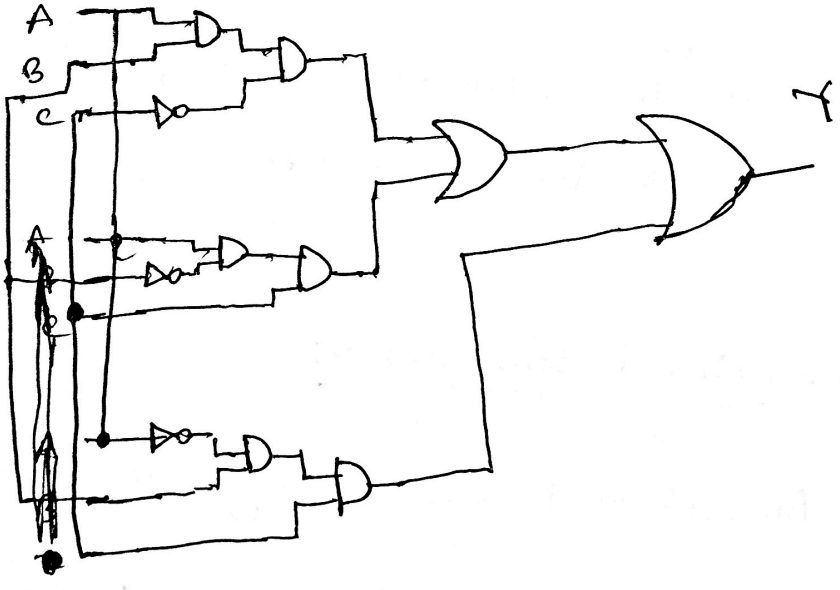
## Required Components:

Bread board, connecting wires, IC 7404 (Not gate), IC 7432 (OR Gate), IC 7408 (AND Gate), Digital Trainer board.

## Theory:

The Boolean expression means the total OR of each AND of the three inputs with one NOTed alternatively.

## Circuit Diagram:



## Procedures:

- Connect A, B, C to NOT gates in order to get $\bar{A}, \bar{B}, \bar{C}$.
- Connect $\bar{A}, B$ into an AND gate and there.
- Repeteadly connect $A\bar{B}C$, $AB\bar{C}$.
- Combine the outputs of them in the OR gate.

## Truth Table:

| A | B | C | $\bar{A}$ | $\bar{B}$ | $\bar{C}$ | $\bar{A}BC$ | $A\bar{B}C$ | $AB\bar{C}$ | Y |
|---|---|---|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 1 | 1 | 1 | 0 | 0 | 0 | 0 |
| 0 | 0 | 1 | 1 | 1 | 0 | 0 | 0 | 0 | 0 |
| 0 | 1 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 |
| 0 | 1 | 1 | 1 | 0 | 0 | 1 | 0 | 0 | 1 |
| 1 | 0 | 0 | 0 | 1 | 1 | 0 | 0 | 0 | 0 |
| 1 | 0 | 1 | 0 | 1 | 0 | 0 | 1 | 0 | 1 |
| 1 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 1 | 1 |
| 1 | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

## Result:

The experimental results matched the theoretical truth table confirming correct implementation.

## Discussion:

The circuit correctly implemented that boolean expression. AND, OR, NOT gate were used. Unused inputs caused unpredictable outputs (fixed them by grounding).